

AI-Assisted Software Development

AI and how it helps Software Development

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Introduction to "AI" and its place in Software Development

- Generative AI has revolutionised many areas of work
 - Including Software Development
- It is becoming an essential tool for developers
- We need to master it!

In this module, we will:

- Introduce Generative AI and Large Language Models
- Examine traditional developer workflows
- Explore AI-powered workflows and their recent progression

Overview of (Generative) "AI"

Why "Generative"?

- Machine Learning gave us Neural Networks to do:
 - Classification
 - Recommendation
 - Regression
- Generative AI **generates**:
 - Text
 - Music
 - Images
 - Videos

Based on Large Language Models

- Built with Transformer architecture
 - From blocks of deep Neural Networks
- Culmination of breakthroughs in:
 - Neural Network techniques
 - Google's "Attention is all you need" paper
 - Hardware
 - **Nvidia**, also:
 - Amazon
 - and even Apple
 - Business
 - Amazon AWS, easy access to hardware
 - Microsoft, "buying" and popularising OpenAI & ChatGPT

What is a Large Language Model (LLM)?

- **Model**
 - Use past data to predict (guess!) the future
 - Simple example: Journey plot
- **Language model**
 - Predict next word
 - Then iterate
- **Large**
 - Several hundred billion parameters (like "factors")
 - Enables a GOOD guess!

Lifecycle of building an LLM

- **Build**
 - Layers of Neural Networks
 - (further reading!)
- **Train**
 - Feed it MANY examples of language(S!)
 - eg entire internet as at a certain date
 - Very compute-intensive
 - Requires thousands of GPUs
 - or other specialised hardware, eg AWS Trainium
- **Fine-Tune**
 - eg Tune for Chat-capability
 - Feed it many Question/Answer pairs
- **Deploy**
 - Locally on BIG hardware
 - GPUs
 - NPUs (Apple A11.. M1.. AWS Inferon)
 - or via API charged per token ("word") sent & received

Using an LLM

- Query

- Send a "prompt"
- Receive a "completion"
- These accumulate into "context" (ie conversation)
- Answer / Reply / Completion is based on available data:
 - Training data
 - Fine-tuning data
 - Context
- Add other data into context
 - Up to "context window" size

- Techniques

- Prompt engineering
- Retrieval Augmented Generation ("RAG")
 - For data bigger than context window
 - Identify a "similar" subset of available data
 - via similarity search using "Vector Database"
 - Add this into the conversation context

Traditional developer workflow

- Familiar path:
 - Easy
 - In context
 - Comfortable
- Unusual path:
 - For out of the ordinary tasks
 - Familiarise, read the docs, read the code
 - (Re-) Learning curve
 - Productive after "warm-up period"
 - Uncomfortable at first (especially if time pressured!)

Early AI developer workflow (c2023)

- ChatGPT
- Perplexity for AI-powered web-searches
- Claude, OpenAI... chat consoles
- Copy and Paste!

2025 AI developer workflow

Integrated IDE tools

- Aware of context
 - Entire codebase or subset
 - Can add specific files, notes, websites...
- Reasoning models can plan
 - Then execute steps of the plan
- Not just coding
 - Also assist other areas of software lifecycle
- Go beyond delivering information
 - Agents can perform actions

2025 AI software development tools

- Basic tools eg Copilot
 - AI powered code completion
- Forefront (at present) is Cursor
 - Understands whole codebase
 - Predicts:
 - code completion "next word"
 - navigation, cursor/location movement
 - editing location
 - Dynamically chooses suitable LLMs
 - Agent capability (including unchecked "YOLO mode")
 - Privacy options available
 - even has special agreement with OpenAI
 - Still need to be aware of privacy re other LLMs
 - eg cursor-tools add-on uses Gemini
- Cursor calms Cursors!

Getting Started

- Simple tools
 - ChatGPT?
 - Perplexity
 - Claude
- Use comprehensive prompts
 - More detail yields better answers
 - Avoid simple Google-style searches
 - Prompts can be several pages long
- Be mindful of
 - Model capability (and cost trade-offs)
 - Context Window size
 - Citation capability
 - Cut-off date
 - Privacy of data and flow (possibly also IPR issues?)
- Advanced tools
 - See later sections